

Arjun Gupta

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EDUCATION

University of Illinois at Urbana-Champaign

PhD in Electrical and Computer Engineering

M.S. in Electrical and Computer Engineering

B.S. in Computer Engineering

- Graduated with Highest Honors (GPA: 3.97)

Urbana, IL

Jan. 2021 - Present

Jan. 2021 - Dec. 2022

Aug. 2017 - Dec. 2020

SELECTED PUBLICATIONS

- | | |
|--|---|
| [1] Precise Mobile Manipulation of Small Everyday Objects
Arjun Gupta , Rishik Sathua, Saurabh Gupta | RA-L 2026
[webpage] |
| [2] Opening Articulated Structures in the Real World
Arjun Gupta , Michelle Zhang*, Rishik Sathua*, Saurabh Gupta | RSS 2025
[webpage] |
| [3] Estimating Perceptual Uncertainty to Predict Robust Motion Plans
Arjun Gupta , Michelle Zhang, Saurabh Gupta | IROS 2024
[webpage] |
| [4] Mitigating Perspective Distortion-induced Shape Ambiguity in Image Crops
Aditya Prakash, Arjun Gupta , Saurabh Gupta | ECCV 2024
[webpage] |
| [5] Predicting Motion Plans for Articulating Everyday Objects
Arjun Gupta , Max E. Shepherd, Saurabh Gupta | ICRA 2023
[webpage] |
| [6] Learning Value Functions from Undirected State-only Experience
Matthew Chang*, Arjun Gupta* , Saurabh Gupta | ICLR 2022
[webpage] |
| [7] Semantic Visual Navigation by Watching YouTube Videos
Matthew Chang, Arjun Gupta , Saurabh Gupta | NeurIPS 2020
[webpage] |

RESEARCH EXPERIENCE

Google

PhD Intern — Manager: Dr. Youbao Tang

- Designed a multimodal VLM framework for long-horizon egocentric video reasoning in robotics.

Sunnyvale, CA

Sept. 2025 - Dec. 2025

Amazon Robotics

Applied Scientist Intern — Manager: Dr. Nicolas Hudson

- Investigated point tracking under occlusion for robot policy learning via sim2real.

Seattle, WA

May. 2025 - Aug. 2025

Hello Robot Inc.

Research Intern — Manager: Dr. Chris Paxton

- Developed general-purpose grasping functionality for Stretch AI using sim2real techniques.

Martinez, CA

Oct. 2024 - Dec. 2024

University of Illinois at Urbana-Champaign

PhD Student — Advisor: Prof. Saurabh Gupta

- **Real World Robots.** Developed end-to-end whole-body mobile manipulation pipelines for real robots (integrating perception, navigation, and manipulation) which generalize to *in-the-wild* settings across 10+ buildings [1, 2].
- **Computer Vision.** Trained state-of-the-art computer vision models for improving robustness in 3D vision [3, 4].
- **Motion Planning.** Developed a TrajOpt-based approach which outperforms existing planning methods [5].
- **Sim2Real RL.** Designed sim2real reinforcement learning approaches for navigation via learning from videos [6, 7].

Urbana, IL

Jan. 2021 - Present

SKILLS

Programming: Python, C/C++, ROS, Matlab.

Software: PyTorch, Keras, Tensorflow, NumPy, SciPy, OpenAI Gym, Git, IsaacGym, MuJoCo.

Hardware: Stretch, Franka, UR, 3D Cameras, F/T Sensors.